# Cloud Computing & AWS Fundamentals

Professional Learning Notes (Fresher–Friendly)

## 1. What is Cloud Computing?

Cloud computing is a technology that enables on-demand access to computing resources such as servers, storage, databases, networking, software, analytics, and intelligence over the internet. Users can consume these resources without owning, purchasing, or maintaining physical infrastructure, following a pay-as-you-go pricing model.

## 2. Cloud Deployment Models

- Public Cloud: Services offered over the internet and managed by third-party providers. Examples include AWS, Microsoft Azure, and Google Cloud Platform.
- Private Cloud: Infrastructure dedicated to a single organization, hosted on-premises or by a third party. Offers higher control and security. Examples include OpenStack, VMware vSphere, and Azure Stack.
- Hybrid Cloud: Combination of public and private clouds allowing data and application portability. Examples include AWS Outposts, Azure Arc, and Google Anthos.
- Community Cloud: Shared infrastructure for organizations with common compliance or security requirements, such as government or healthcare sectors.
- Multi-Cloud: Use of multiple cloud providers to avoid vendor lock-in and optimize performance, such as using AWS for compute, Azure for AI, and GCP for analytics.

## 3. Cloud Service Models

- Infrastructure as a Service (IaaS): Provides virtualized computing resources like servers, storage, and networking. Example: AWS EC2.
- Platform as a Service (PaaS): Provides a managed platform with OS, runtime, and middleware for application deployment. Example: AWS Elastic Beanstalk.
- Software as a Service (SaaS): Delivers ready-to-use software applications over the internet. Example: Google Workspace, Microsoft 365.

## 4. Virtualization Fundamentals

Virtualization allows a single physical server to run multiple virtual machines (VMs), each with its own operating system and allocated resources. This improves hardware utilization, flexibility, and cost efficiency.

### Hypervisor

A hypervisor is the software layer that manages virtual machines and allocates physical resources such as CPU, memory, storage, and networking.

- Type 1 (Bare Metal): Runs directly on hardware. High performance. Example: VMware ESXi, AWS EC2.
- Type 2 (Hosted): Runs on top of an existing OS. Example: VirtualBox, VMware Player.

### Containerization & Serverless

Containers provide lightweight virtualization by sharing the host OS kernel. Serverless computing removes server management entirely, allowing developers to run code on demand.

## 5. Amazon Web Services (AWS)

Amazon Web Services (AWS) is a global cloud platform offering on-demand services with a pay-as-you-go pricing model. It is a subsidiary of Amazon.com.

- IAM – Identity and Access Management
- EC2 – Elastic Compute Cloud
- S3 – Simple Storage Service
- RDS – Relational Database Service
- Lambda – Serverless Compute
- VPC – Virtual Private Cloud
- CloudFront – Content Delivery Network
- Route 53 – DNS Service
- CloudWatch – Monitoring and Logging

## 6. Amazon EC2 (Elastic Compute Cloud)

Amazon EC2 provides scalable virtual servers in the cloud. Users can launch, stop, and scale instances based on workload demands.

- Instance – Virtual server
- AMI – Amazon Machine Image
- EBS – Elastic Block Store
- Security Groups – Virtual firewall
- Key Pairs – Secure login
- Elastic IP – Static public IP
- Load Balancer – Traffic distribution
- Auto Scaling Group – Automatic scaling

## EC2 Instance Lifecycle

- Launching
- Running
- Stopping
- Rebooting
- Terminating